MIL-C-37586
29 September 1977

MILITARY SPECIFICATION

CAP, STERILE FLUID STORAGE BOTTLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable rubber closure, suitable for vacuum sealing of sterile fluid storage bottles.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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MILITARY

MIL-P-116	- Preservation-packaging, Methods of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish a pilot lot, consisting of at least 3 caps from each mold cavity to be used in the production of the caps, for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Material. Material used in the caps shall be rubber suitable for the purpose intended, and capable of meeting the requirements of the tests specified in section 4.4.

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3.3 Design and construction. The cap shall be of single piece construction. It shall include a pull-off tab, an internal air-escape vent, a means of maintaining a vacuum-seal, and a vacuum-seal indication when fitted onto a bottle with the mouth-configuration illustrated in figure 1. The cap shall be designed to meet the following performance criteria. During sterilization, the cap shall permit free passage of air and steam vapor through the escape vent. A noticeable depression shall form in the center of the closure after the bottle solutions have cooled, indicating a positive vacuum seal. The cap shall resume its normal contour, automatically, to indicate that the seal has been broken. The cap shall perform as specified above, when tested in accordance with 4.4.1. There shall be no evidence of bacterial growth when tested as specified in 4.4.1. The overall dimensions of the caps shall be not more than 3 inches in diameter (excluding the tab), nor 3/4 inch in height. The top of the cap shall be clearly imprinted with the word, "Disposable", as a minimum.

3.4 Toxicity (safety). Shall be non-toxic when tested as specified in 4.4.2.

3.5 Pyrogenicity. Shall be non-pyrogenic when tested as specified in 4.4.3.

3.6 PH effect. The pH of the solution shall be within the limits of 4.5 to 7.0 when tested as specified in 4.4.4.

3.7 Heavy metals. The color of the solution shall be no darker than the color of the non-sterilized distilled water and acetic acid solution when tested as specified in 4.4.5.

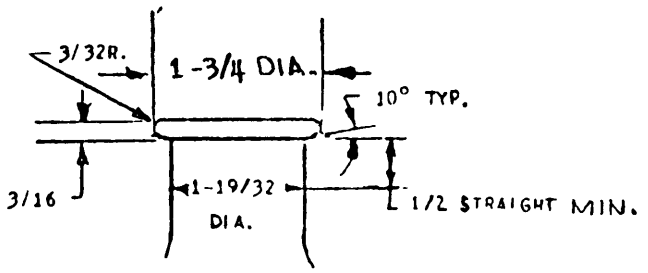
3.8 Oxidizable material. The color shall not disappear completely from the solution when tested as specified in 4.4.6.

3.9 Foreign matter. No foreign matter shall be observed in the solution when tested as specified in 4.4.7.

3.10 Extractable matter. No extractable matter shall be observed in the solution when tested as specified in 4.4.8.

3.11 Identification marking. Each unit shall be marked permanently and legibly with the contract or purchase order number, or the name or registered trademark of the manufacturer.

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TOLERANCES UNLESS OTHERWISE SPECIFIED
FRACTIONS ±1/32
ANGLES ±2°

NOTE: ALL LINEAR DIMENSIONS ARE IN INCHES

FIGURE 1. Mouth of Sterile Fluid Storage Bottle.

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3.12 Process control. Manufacturer shall perform in-process inspection on a periodic basis to ensure maintenance of quality standards.

3.13 Workmanship. Workmanship shall be first class throughout. Cap shall be free from defects which detract from its appearance or impair its serviceability. No molding flash shall be present on the cap. Cap shall be of uniform color and shall have a smooth surface with no voids, cuts, tears, or irregularities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the cap when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I. The unit of product for sampling purposes shall be one cap.

TABLE I. Sampling for examination.

	Inspection level		AQL (percent defective)
	First Article	Quality Conformance	
For visual examination			
Major defects	100%	I	1.0
Minor defects	100%	I	2.5
For dimensional examination	1/	1/	1.0

1/ One cap from each mold cavity.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be one cap. The acceptance number shall be zero for all sample sizes. In the first article sterility retention test, should a cap from any mold cavity fail to meet the requirements, two other caps from the same cavity shall be tested. If both meet the requirements, the cavity shall be considered to be acceptable.

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TABLE II. Sampling for tests.

Characteristic	Require-ment	Test procedure	Inspection level	
			First Article	Quality Conformance
Sterility retention	3.3	4.4.1	1/	S-3
Toxicity (Safety)	3.4	4.4.2	S-4	S-3
Pyrogenicity	3.5	4.4.3	S-4	S-3
pH	3.6	4.4.4	S-4	S-3
Heavy metals	3.7	4.4.5	S-4	S-3
Oxidizable material	3.8	4.4.6	S-4	S-3
Foreign matter	3.9	4.4.7	S-4	S-3
Extractable matter	3.10	4.4.8	S-4	S-3

1/ One cap from each mold cavity.

4.3 Examination. The caps shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Cap not complete - free of voids (incomplete mold fill).
102	Cap has flash or other structural imperfections.
103	Cap does not contain clear, legible directions and identification marking as specified.
104	Cap is not supplied clean and dry.
Minor	
201	Cap has non-uniform color or appearance.

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4.3.2 Dimensional examination. The caps shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation - packaging and interior package marking shall be in accordance with the group A quality conformance inspection requirements of MIL-P-116. The inspection of the packaging and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterility retention test. Sterility retention test shall be performed as follows:

a. Take caps, wash in soap and water, then rinse in distilled water, and dry.

b. Soak caps in distilled water for 2 minutes.

c. Take 500cc solution bottles, conforming to figure 1, and place 250cc of soybean-casein digest medium (USP) in each bottle.

d. Snap caps onto bottles, rotating caps 1/4 turn.

e. Sterilize the solutions for 30 minutes at a maximum temperature of 121° C in a properly-cycled sterilizer. NOTE: for purposes of this specification, a properly-cycled sterilizer is one that is capable of sterilizing solutions according to standard accepted practice. The principle requirements are that the chamber reaches 121° C within approximately 10 minutes after steam is introduced and exhausts after the exposure time at 121° C at a slow enough rate to prevent explosion or boiling over of the fluid from the flask (approximately 15 minutes exhaust time).

f. After sterilization, remove bottles from sterilizer and allow to cool to room temperature. Inspect cap for depression.

g. Allow sterilized bottles to stand at 20° C to 25° C and inspect for bacterial growth after 24 hours, 48 hours, and every 7 days thereafter for an inclusive total of 30 days.

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4.4.2 Toxicity (safety) test. Fill a clean 500 ml sterile fluid storage bottle with a solution of 0.9 percent W/V Sodium Chloride in freshly distilled water, insert one-half of a cleaned cap into the flask with the saline solution, apply a clean cap freshly rinsed in distilled water, and expose to 121° C for 60 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. In accordance with the USP Biological Test for Transfusion and Infusion assemblies, 0.5 ml of solution shall be injected into each of five healthy mice. The mice shall be examined after 4, 24, and 48 hours for death or symptoms of toxicity.

4.4.3 Pyrogen test. Fill a clean 500 ml sterile fluid storage bottle with a solution of 0.9 percent W/V Sodium Chloride in freshly distilled water, insert one-half of a clean cap into the flask with the saline solution, apply a clean cap freshly rinsed in distilled water, and expose to 121° C for 60 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. In accordance with the USP Biological Test for Pyrogens, the solution shall be injected into rabbits and the temperature of the rabbits shall be checked.

4.4.4 pH test. Fill a clean 500 ml sterile fluid storage bottle with a solution of 0.9 percent W/V Sodium Chloride in freshly distilled water, insert one-half of a cleaned cap into the flask with the saline solution, apply a clean cap freshly rinsed in distilled water, and expose to 121° C 60 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. Determine the pH using the USP method.

4.4.5 Heavy metals test. Fill a clean 500 ml sterile fluid storage bottle with freshly distilled water and apply a clean cap freshly rinsed in distilled water. Expose to 121° C for 30 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. Place the sealed flask in a hot air oven for 24 hours at 82° C. Cool to room temperature. Add 1 ml of 1 N Acetic Acid and 10 ml of a saturated water solution of Hydrogen Sulfide to 40 ml of the distilled water in a long Nessler color comparison tube; after 10 minutes the color of the liquid shall be compared to the color of 50 ml of non-sterilized freshly distilled water, to which has been added 1 ml of 1 N Acetic Acid, in a matched tube, viewing downward over a light surface.

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4.4.6 Oxidizable material test. Fill a clean 500 ml sterile fluid storage bottle with freshly distilled water and apply a clean cap freshly rinsed in distilled water. Expose to 121° C for 30 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. Place the sealed flask in a hot air oven for 24 hours at 82° C. Cool to room temperature. To 100 ml of the distilled water, add 10 ml of 10 percent W/V (5.7 percent V/V) Sulphuric Acid, heat to boiling; then add 0.1 ml of 0.1 N Potassium Permanganate. Check for the presence of color after 30 minutes reflux boiling.

4.4.7 Foreign matter examination. Fill a clean 500 ml sterile fluid storage bottle with freshly distilled water and apply a clean cap freshly rinsed in distilled water. Expose to 121° C for 30 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. Place the sealed flask in a hot air oven for 24 hours at 82° C. Cool to room temperature. Check for the presence of foreign matter (such as loose stock, dust, talc, etc.) in the solution, as determined by observation, aided by a strong beam of white light passed through the solution in a darkened room.

4.4.8 Extractable matter examination. Fill a clean 500 ml sterile fluid storage bottle with freshly distilled water and apply a clean cap freshly rinsed in distilled water. Expose to 121° C for 30 minutes in a properly cycled sterilizer (see 4.4.1). Cool to room temperature. Place the sealed flask in a hot air oven for 24 hours at 82° C. Cool to room temperature. Check for the presence of extractable matter (such as bloom, haze, color, etc.) in the solution, as determined by observation, aided by a strong beam of white light passed through the solution in a darkened room.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing 100 caps as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Cleaning and drying. Caps shall be cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.1.2 Unit packaging. One hundred caps shall be packaged in a plastic bag of suitable size. Closure of the bag shall be by folding and securing by ties or other suitable means (Method III, MIL-P-116).

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5.1.1.2 Level C. One hundred caps shall be packaged in standard commercial containers of the size and kind commonly used, which will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Level A. Ten (10) unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3 , style 3; PPP-B-601 overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Ten (10) unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged caps shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. The container shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The caps are intended for use with bottles, sterile fluid storage, NSN 6530-00-935-4136 (1,000 cc) and NSN 6530-00-935-4137 (500 cc).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3) and the number of caps per mold cavity to be included in the first article quantit
d. Place of performance of first article testing.

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-113-3724	CAP, STERILE FLUID STORAGE BOTTLE, Disposable, 100s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6530-1412

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37586 CAP, STERILE FLUID STORAGE BOTTLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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